25901053 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCREM102 - MATHEMATICS FOR RENEWABLE ENERGY- I

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **15** questions

Each question carries **1** marks

1. Find the HCF of 8 and 12 [K] / [CO1]
2. Simplify: (5/6) ÷ (10/9) [K] / [CO1]
3. What is the rank of a matrix? [A] / [CO3]
4. What is the condition for the multiplication of two matrices to be possible? [A] / [CO3]
5. Give an Example for Lower Triangular Martrix of order 3 x 3 [A] / [CO3]
6. What is the determinant of $\begin{bmatrix} 6 & 4 \\ 3 & 7 \end{bmatrix}$ [A] / [CO3]
7. What is the order of a 2 × 3 matrix? [A] / [CO3]
8. Define scalar multiplication of a matrix. [An] / [CO3]
9. What is the Value $Sin60^{\circ}$ [An] / [CO4]
10. Define $Tan\ \theta$ in terms of $Sin\ \theta\ and\ Cos\theta$ [An] / [CO4]

11. A set with no elements in it is called? [U] / [CO5]

12. What is the symmetric difference of sets A and B [U] / [CO5]

13. Identify if the set of positive integers is finite or infinite [U] / [CO5]

14. State the idempotent law for intersection [U] / [CO5]

15. Find the power set of {1, 2}. [U] / [CO5]

16. What does "∪" stand for in set theory? [U] / [CO5]

17. If A-B=B-A then A and B are _____ [U] / [CO5]

[1x15 = 15]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

18. Write any three properties of identity matrix. [A] / [CO3]

19. Show that $(A^{-1}) = I$ with an example. [An] / [CO3]

20. Find the period and frequency of $y = \cos(2x + 45^\circ)$ [An] / [CO4]

21. Derive the general equation of circle from its standard form. [An] / [CO4]

22. Explain the difference between roster form and tabular form of sets with examples. [U] / [CO5]

[5x3 = 15]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

23. Simplify the following expression step by step and find the final answer [K] / [CO1]

$$\left[5 \times (2^2 + 3) - \sqrt{49}\right] \div 3 + \left[16 \div 4 + (9 - 2^2) \times 2\right] - \sqrt{81}$$

24. For matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$ $verify\ that\ (AB)^T = B^T A^T$. [A] / [CO3]

25. For the function $y = 5\sin(2x + 45^\circ)$, determine amplitude, frequency, and phase shift, and Time Period [An] / [CO4]

26. State and prove distributive laws of union and intersection of sets with examples. [U] / [CO5]

[10x2 = 20]